

**EXPT
NO:08**

MINI PROJECT -Airline Passenger Satisfaction Analytics

AIM:

To analyze airline passenger satisfaction data using Python, SQL Server, and Tableau by preparing and managing a 20,000-record dataset, performing data analysis, and creating interactive visualizations to identify trends in passenger satisfaction, class, flight distance, customer type, and service ratings.

Software and Tools Required

- Python
- VS Code
- SQL Server
- SQL Server Management Studio (SSMS)
- Tableau Desktop

3. Process / Procedure

Step 1 — Project Setup

Create a project folder:

Mini Project - 8

```
|— train.csv
|— prepare_data.py
|— airline_passenger_20000.csv
```

Create and prepare the Python project.

Step 2 — Create Database

Open SQL Server Management Studio (SSMS) and connect using:

Server: localhost

Authentication: Windows Authentication

Create the database:

```
CREATE DATABASE airline_analytics;
GO
```

Step 3 — Create Table

Create the airline_passenger_data table:

```
USE airline_analytics;
GO
```

```
CREATE TABLE airline_passenger_data (
    Passenger_ID INT PRIMARY KEY,
    Gender VARCHAR(20),
    Customer_Type VARCHAR(50),
```

```
Age INT,  
Type_of_Travel VARCHAR(50),  
Class VARCHAR(30),  
Flight_Distance INT,  
Inflight_Wifi_Service INT,  
Departure_Arrival_Time_Convenient INT,  
Ease_of_Online_Booking INT,  
Gate_Location INT,  
Food_and_Drink INT,  
Online_Boarding INT,  
Seat_Comfort INT,  
Inflight_Entertainment INT,  
On_Board_Service INT,  
Leg_Room_Service INT,  
Baggage_Handling INT,  
Checkin_Service INT,  
Inflight_Service INT,  
Cleanliness INT,  
Departure_Delay INT,  
Arrival_Delay INT,  
Satisfaction VARCHAR(50)  
);  
GO
```

Step 4 — Generate Dataset Using Python

Python is used to prepare exactly 20,000 airline passenger records.

The dataset contains:

- Passenger ID
- Gender
- Customer Type
- Age
- Type of Travel
- Class
- Flight Distance
- Inflight Wifi Service
- Departure/Arrival Time Convenient
- Ease of Online Booking
- Gate Location
- Food and Drink
- Online Boarding
- Seat Comfort
- Inflight Entertainment
- On-board Service

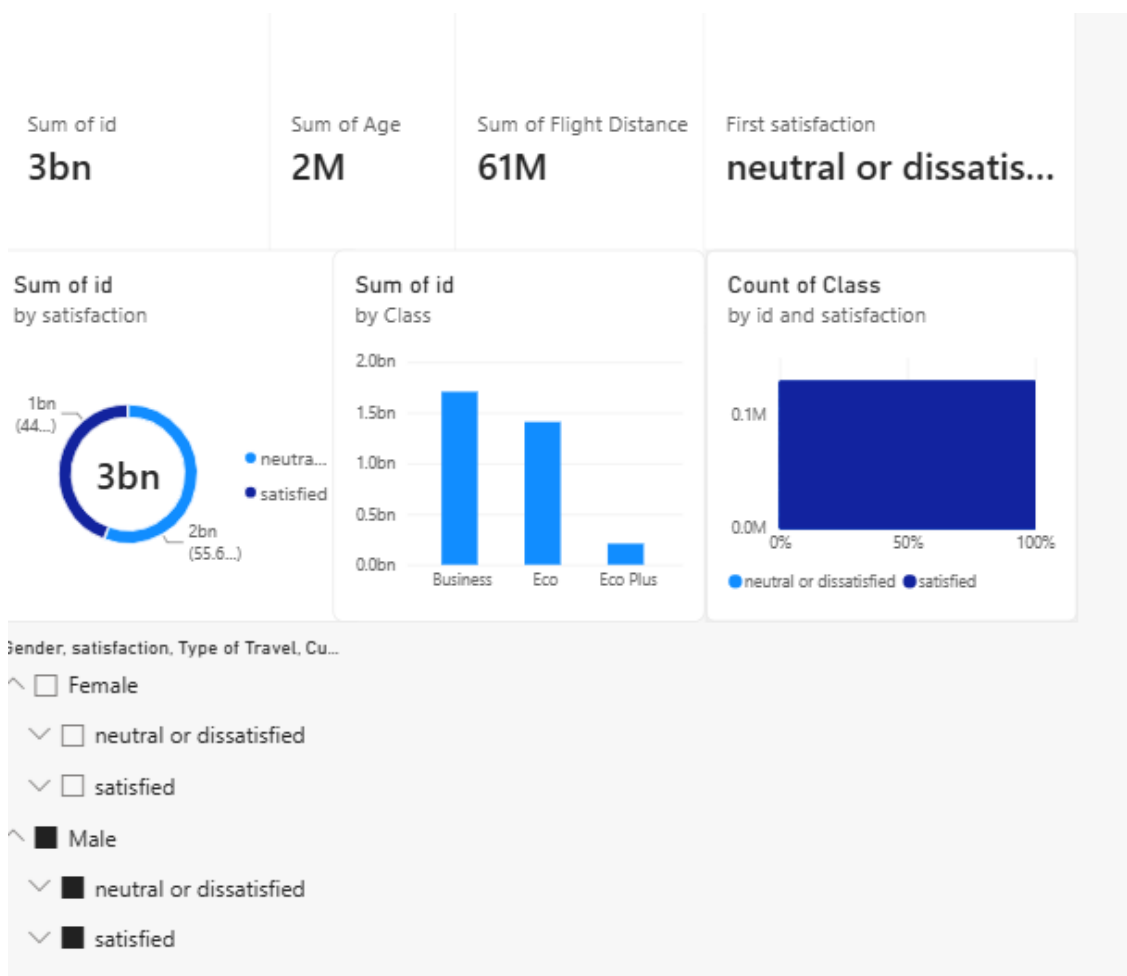
- Leg Room Service
- Baggage Handling
- Checkin Service
- Inflight Service
- Cleanliness
- Departure Delay
- Arrival Delay
- Satisfaction

The prepared data is saved as:

airline_passenger_20000.csv

OUTPUT:

Python Data Processing



The dashboard screenshot contains the passenger satisfaction, class, and satisfaction-by-class visualizations.

Step 5 — Insert Data into SQL Server

The prepared dataset is imported into SQL Server and stored in the airline_analytics database.

The data flow is:

Python

↓
airline_passenger_20000.csv

↓
SQL Server

↓
airline_analytics

↓
airline_passenger_data

Step 6 — Verify the Dataset

Run:

```
USE airline_analytics;
```

```
GO
```

```
SELECT COUNT(*) AS total_records
```

```
FROM airline_passenger_data;
```

Result:

```
total_records
```

```
-----
```

```
20000
```

This confirms that all 20,000 records have been successfully stored in SQL Server.

4. Data Visualization Using Power Bi

After storing the dataset in SQL Server, connect Power Bi Desktop to:

SQL Server

↓
airline_analytics

↓
airline_passenger_data

Create different visualizations.

Visualization 1 — Passenger Satisfaction

Chart: Donut Chart

Shows the distribution of passengers based on satisfaction.

Visualization 2 — Passengers by Class

Chart: Bar Chart

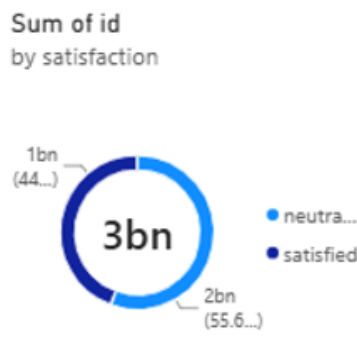
Shows the passenger distribution across different classes.



Visualization 3 — Satisfaction by Class

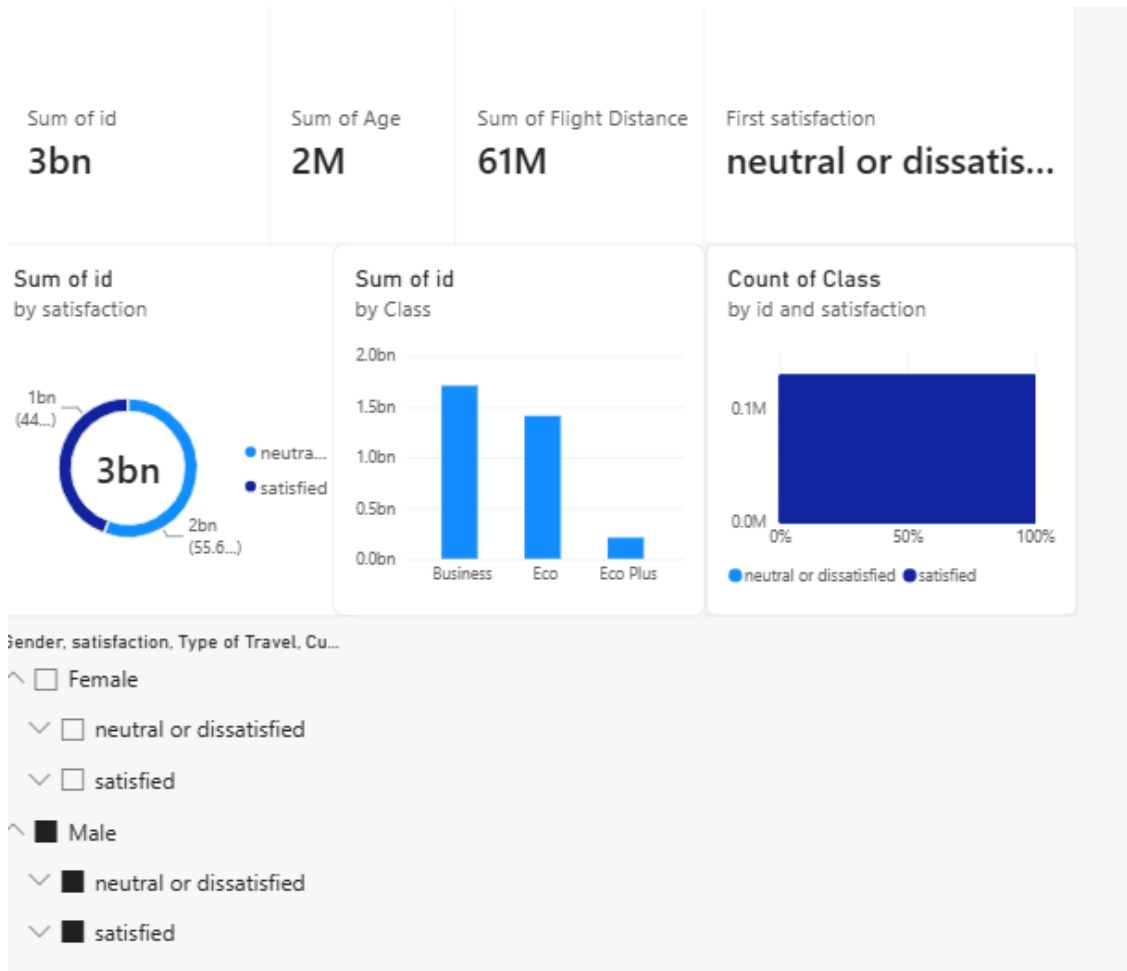
Chart: Bar Chart

Shows satisfaction across different passenger classes.



5. Dashboard

Combine the visualizations into a single interactive dashboard.



The dashboard can include filters for:

- Gender
- Customer Type
- Type of Travel
- Class
- Satisfaction

RESULT:

The Airline Passenger Satisfaction mini project was successfully implemented using Python, SQL Server, and Tableau. A dataset containing exactly 20,000 airline passenger records was prepared using Python and stored in SQL Server. The data was successfully analyzed and visualized using Tableau through bar and donut charts. The dashboard provides insights into passenger satisfaction, class, and airline passenger characteristics.

Final Result

20,000 Records Prepared

↓

Python Data Processing

↓

CSV Dataset Created

↓
SQL Server Database
↓
20,000 Records Verified
↓
Tableau Connection
↓
Charts & Visualizations
↓
Interactive Dashboard
↓
Airline Passenger Satisfaction Analytics Completed

CONCLUSION:

Thus, the airline passenger dataset was successfully prepared, stored, managed, analyzed, and visualized using Python, SQL Server, and Tableau. The project demonstrates how data analytics and visualization techniques can be used to understand passenger satisfaction and identify meaningful patterns in airline data.